

Outline

- Input Devices
- Output Devices

What are Input Devices?

- Input devices are hardware components that allow users to input data into a computer or control the system.
- These devices enable communication between the user and the machine, making it possible to interact with the system.

Examples of Input Devices:

- ✓ **Keyboard:** Used to input text and commands by pressing keys.
- ✓ **Mouse:** A pointing device that allows users to interact with the graphical interface of a computer.
- ✓ **Scanner:** Converts physical documents into digital form by scanning them.
- ✓ **Microphone:** Captures audio signals and converts them into digital data.
- ✓ **Webcam:** Captures video and images, which are processed by the computer.

What are Output Devices?

- Output devices are hardware components that **display, print, or otherwise output** data from the computer to the user.
- After the computer processes the input data, the results are sent to output devices for display or further action.

Examples of Output Devices:

- ✓ **Monitor:** Displays visual output, such as text, images, and videos, generated by the computer.
- ✓ **Printer:** Produces hard copies of documents and images on paper.
- ✓ **Speakers:** Output audio signals generated by the computer.
- ✓ **Projector:** Projects the computer screen onto a larger surface for presentations.

How I/O Devices Work with the Computer

1. Data Input:

1. The input device sends signals (e.g., keystrokes, mouse movements) to the computer.
2. These signals are converted into a form that the computer can understand through a process called **signal processing**.
3. The computer processes the input data based on instructions from the operating system or software.

2. Data Output:

1. Once the computer processes the input, it sends the results (e.g., text, images, or sound) to an output device.
2. The output device then converts the processed digital data into a format that the user can perceive (e.g., displaying text on a monitor or playing audio through speakers).

Types of Input Devices

1. Text Input Devices

Keyboard:

- Types: QWERTY, Dvorak, Colemak layouts.
- Key functions: Letters, numbers, special characters, control keys (Shift, Enter, etc.).

Touchscreen Keyboard:

- Used in mobile phones, tablets, and other touchscreen devices.

2. Pointing Devices

Mouse:

- Left-click, right-click, scroll wheel.

Trackpad/Touchpad:

- Common in laptops.
- Gestures: tapping, scrolling, zooming.

Stylus & Graphics Tablet:

- Used for drawing, precise touch input in digital art and design.

3. Audio Input Devices

Microphone:

- Used for voice input, recording, communication.

Voice Recognition Systems:

- Examples: Siri, Google Assistant.

Types of Input Devices cont..

4. Visual Input Devices

Scanner:

- Converts physical documents into digital form.
- Applications in document management and image digitization.

Webcam:

- Captures video for communication, streaming, or video conferences.

Digital Camera:

- Used for capturing high-quality photos and videos.

5. Biometric Input Devices

Fingerprint Scanner:

- Used for authentication.

Face Recognition:

- Security features in mobile phones, laptops, and access control systems.

6. Game and Control Input Devices

Joystick & Gamepad:

- Common in gaming, used for movement and control.

Steering Wheel:

- Used for racing games and simulators.

Types of Input Devices cont..

7. Reading Devices :

- Barcode Scanner
- QR Code Reader
- Card Reader
- Optical Character Recognition (OCR) Devices
- Fingerprint Scanner
- Magnetic Stripe Reader

Types of Keyboards

- ✓ **QWERTY:** The most common keyboard layout, primarily used in English-speaking countries. The name comes from the first six letters in the top row.
- ✓ **Dvorak:** An alternative to QWERTY, designed to increase typing speed by placing the most commonly used keys under the strongest fingers. It has a more logical layout.
- ✓ **Colemak:** Another alternative to QWERTY, designed to be a compromise between QWERTY and Dvorak, offering more efficiency while keeping many of the familiar key placements.
- ✓ **AZERTY:** Used mainly in France and Belgium, it differs from QWERTY in the placement of some letters, like swapping A and Q, and other special characters.
- ✓ **QWERTZ:** Used in German-speaking countries. It has the Z and Y keys swapped compared to QWERTY.
- ✓ **Numeric Keypads:** Keypads focused on numbers, often found on calculators, phones, and some computer keyboards. They usually follow a 3x3 grid with an enter key.
- ✓ **Ergonomic Keyboards:** These keyboards are designed to reduce strain on the hands and wrists, with split key arrangements or curved shapes.

Input Devices – QWERTY Keyboard

- The QWERTY keyboard layout is the most commonly used keyboard layout worldwide.
- It was designed in the 1870s by Christopher Sholes, the inventor of the typewriter, and was originally intended to prevent jamming of mechanical typewriters by placing commonly used letters apart from each other.
- The layout features a staggered arrangement of keys, with letters arranged in the following order on the top row: Q-W-E-R-T-Y.
- Despite its widespread adoption, the QWERTY layout has been criticized for its inefficiency and discomfort, as many commonly used letters and combinations are not optimized for finger movement.

Input Devices – Dvorak Keyboard

- The Dvorak Simplified Keyboard, commonly referred to as the Dvorak keyboard, is an alternative keyboard layout designed for increased typing speed and comfort.
- It was patented in 1936 by Dr. August Dvorak and his brother-in-law, Dr. William Dealey, who aimed to create a more efficient layout by placing the most commonly used letters in the English language on the home row for easier access.
- The Dvorak layout rearranges the keys from the QWERTY layout significantly.
- It places the most commonly used letters on the home row (middle row), with vowels on the left-hand side and consonants on the right-hand side.
- The Dvorak layout is designed to minimize finger movement, reduce strain, and increase typing speed and accuracy.
- It is claimed to be more efficient than the QWERTY layout for touch typing.

QWERTY

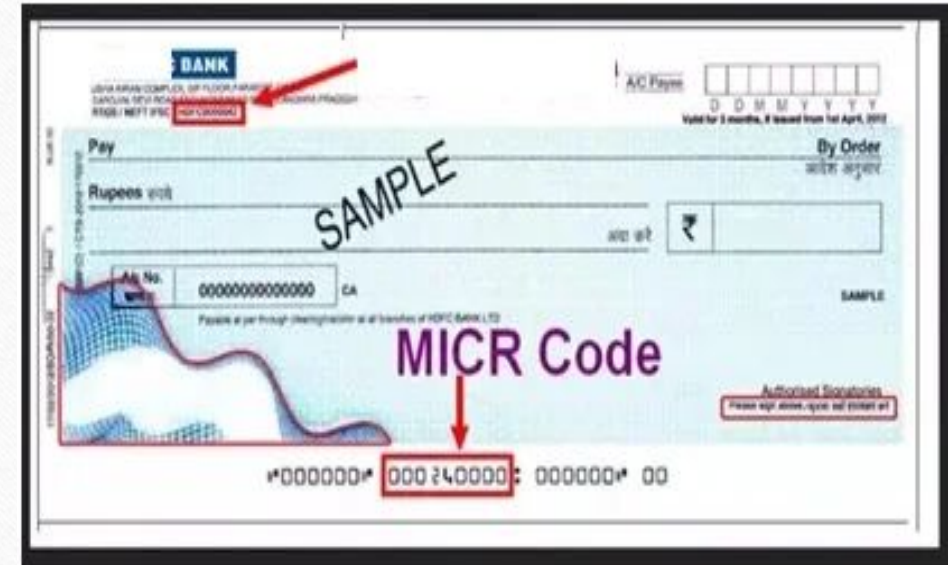


Reading Devices

- A character reader is capable of accepting printed or typed characters from source documents and converting this data into a computer acceptable code.
- The basic types of character readers are :
 - Magnetic ink character reader (MICR)
 - Optical mark reader (OMR)
 - Optical character reader (OCR)
 - Handwritten character reader
 - Smart card etc.....

Magnetic Ink Character Reader (MICR)

- A Magnetic Ink Character Reader (MICR) is a technology used primarily in banking to process checks and other financial documents.
- It involves printing characters using magnetic ink which can be easily recognized by machines equipped with magnetic reading heads.
- The characters printed with magnetic ink contain information such as the bank's routing number, the account number, the check number etc.
- MICR readers use magnetic sensors to detect and decode these characters, allowing for automated processing of checks in banking systems.



Optical Mark Reader (OMR)

- An Optical Mark Reader (OMR) is a technology used to detect human-marked data on documents such as surveys, tests, and forms. OMR systems are commonly used in various fields including education, market research, elections, and data collection.
- Forms are designed with areas where respondents can mark their responses using a pen or pencil, typically using checkboxes, bubbles, or lines to fill in.
- The forms are scanned using an OMR scanner or reader.
- This scanner uses a light source and sensors to detect marks made on the paper.
- The scanned images are then processed using OMR software.
- This software analyzes the images, detects the presence of marks in specific areas, and translates those marks into digital data.

Optical Mark Reader (OMR) cont..

OMR ANSWER SHEET
Presented by **antargyan**

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Roll Number

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Optical Character Recognition (OCR)

- Devices that scan printed or handwritten text and convert it into editable digital text.
- Common in document scanning and digitization tasks.



Difference between OMR and OCR

OMR (Optical Mark Reader) :

Function: OMR is designed to detect and interpret human-marked data, such as checkboxes, bubbles, or lines filled in on paper forms.

Data Processed: OMR systems are primarily used to process structured data where respondents indicate their choices by marking predefined areas on a form.

Examples of Use: OMR is commonly used in surveys, tests, questionnaires, and other forms where respondents need to indicate their choices by marking specific areas.

OCR (Optical Character Recognition) :

Function: OCR is designed to recognize and convert printed or handwritten text characters into machine-readable and editable text.

Data Processed: OCR systems are used to process unstructured data containing text, such as scanned documents, PDF files, or images captured by digital cameras.

Examples of Use: OCR is commonly used to digitize printed documents, automate data entry from paper forms, extract text from images for translation or analysis, and enable text search in scanned documents.

Pointing Devices

- Computer users frequently find it easier to point to something on a screen or at an item of text or graphical material they are entering into the computer.
- A number of such devices are available :
 - Mouse
 - Light Pen
 - Digitizer Pad
 - Joystick and Trackball
 - Touchscreen

Pointing Devices



Scanning Devices

- Scanner
- Barcode Reader
- Digital Camera

Scanner

- Scanners allow a PC to convert a drawing or photograph into code that a graphics or desktop publishing program can use to display the image on the screen to reproduce the image with a printer.
- It can also convert printed text into editable text with the help of optical character recognition software.
- **Three basic types of scanners are :**
 - Sheet-feed Scanner
 - Flatbed Scanner and
 - Hand-hold Scanner

Scanner – (Sheet-feed Scanner)

- A sheet-feed scanner, also known as an automatic document feeder (ADF) scanner, is a type of scanner that automatically feeds documents into the scanning mechanism.
- This is particularly useful for scanning multiple documents or pages in quick succession without the need for manual intervention between each scan.
- Sheet-feed scanners are commonly used in office environments or for tasks requiring the scanning of multiple documents, such as digitizing receipts, contracts, or forms.



Scanner – (Flatbed Scanner)

- A flatbed scanner is a type of scanner that consists of a flat, glass surface on which documents, photos, or other flat objects are placed for scanning.
- Flatbed scanners are versatile and can handle a wide range of materials, including documents, photographs, books, magazines, and even 3D objects (when the scanner lid can be raised to accommodate thicker items). artwork, and more.



Scanner – (Handhold Scanner)

- A handheld scanner is a portable device that allows users to scan documents, images, or barcodes by manually moving the scanner over the surface of the item being scanned.
- These scanners are compact, lightweight, and designed for on-the-go scanning tasks where mobility is essential.



Output Devices

- Output devices in a computer are peripherals that receive data from the computer and present it to the user or another system.
- Some common output devices in a computer system include :
 - Monitor
 - Printer
 - Projector
 - Speaker
 - Plotter etc...

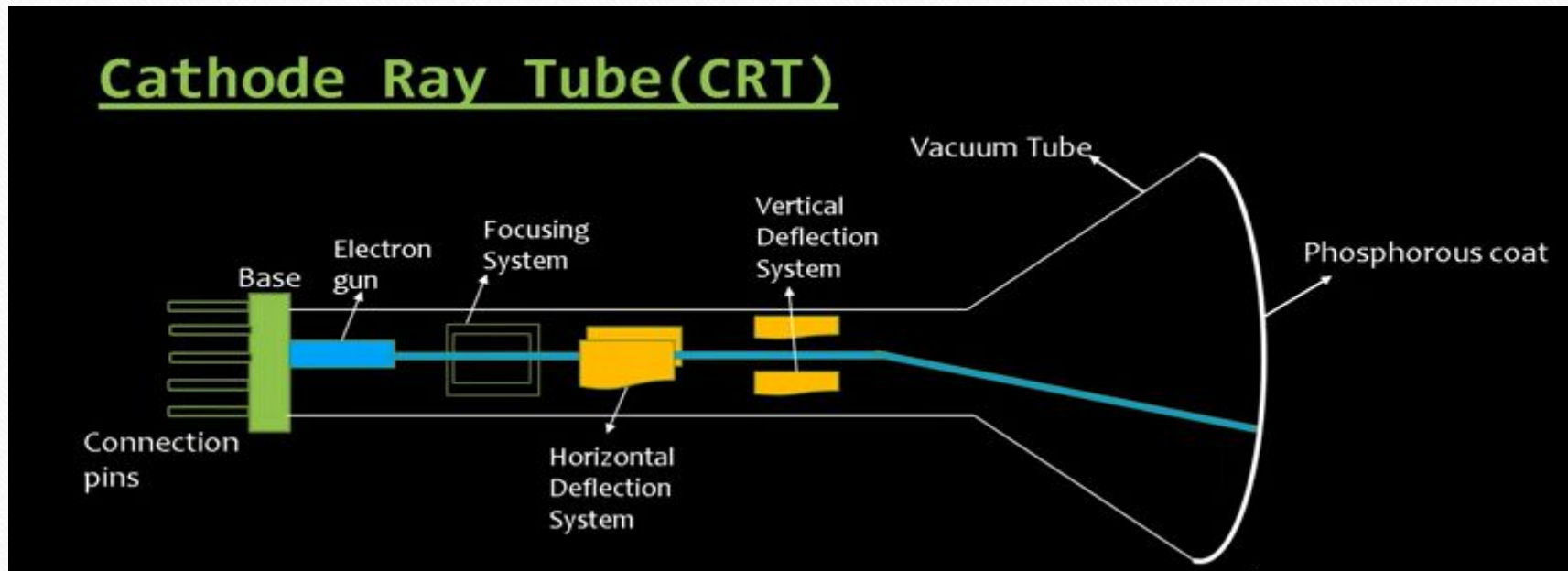
Monitor

- Monitors, commonly called as Visual Display Unit (VDU), are the main output device of a computer.
- It forms images from tiny dots, called pixels (**P**icture **E**lement) that are arranged in a rectangular form.
- The sharpness of the image depends upon the number of pixels.
- There are two kinds of viewing screen used for monitors.
 - Cathode-Ray Tube (CRT)
 - Flat-Panel Display

Cathode Ray Tube (CRT) Monitor Working Principle

- **Electron Gun** : At the back of the CRT, there is an electron gun which emits a narrow beam of electrons.
- **Electron Beam Formation** : The electron gun generates a stream of electrons when a high voltage is applied to the cathode.
- **Deflection System** : Once the electron beam is formed, it passes through a series of electromagnets or electrostatic plates that can deflect the beam both horizontally and vertically.
- **Phosphorescent Screen** : The front of the CRT is coated with a phosphorescent material. When struck by the electrons, this material emits light.
- **Image Formation** : By varying the intensity of the electron beam and controlling its horizontal and vertical movements through the deflection system, the electron beam "paints" the desired image on the screen.

Cathode Ray Tube (CRT) Monitor Working Principle cont..



Flat Panel Displays

- Flat panel displays (FPDs) are modern display screens that are thin, lightweight, and use different technologies to produce images.
- These displays are commonly found in televisions, computer monitors, smartphones, and tablets.
- There are several types of flat-panel display technologies, each with its own characteristics and use cases.

Display Type	Backlight Technology	Best Use Case	Pros	Cons
LCD	CCFL or LED	General use, budget monitors	Cost-effective, widely available	Limited color range, viewing angles
LED	LED	TVs, gaming monitors, laptops	Slim, energy-efficient, brighter displays	Edge-lit models can have uneven brightness
OLED	Organic light emission	High-end TVs, gaming, professional work	Superior contrast, true blacks, better color	Expensive, potential burn-in issues
Plasma	Plasma technology	Large TV screens, home theaters	Great black levels, vibrant colors	High power consumption, bulky, discontinued
TFT	Thin film transistors	Mobile devices, laptops	Sharper images, compact, energy-efficient	Narrow viewing angles
IPS	In-plane switching	Design, photography, professional use	Better color and viewing angles	Expensive, slower response times
TN	Twisted nematic	Gaming, budget monitors	Fast response time, inexpensive	Poor color accuracy, narrow viewing angles
MicroLED	Micro-LED	High-end displays, scalable large screens	Excellent color and contrast, no burn-in issues	Expensive, still emerging

What is Resolution in a Monitor?

- ✓ Resolution refers to the number of pixels displayed on a monitor's screen.
- ✓ Pixels are the smallest units of a digital image, and the resolution is expressed as the number of pixels in the horizontal and vertical dimensions.

How Resolution is Expressed?

- ✓ Monitor resolution is written as:Width x Height
- ✓ Example: 1920 x 1080 (1920 pixels horizontally and 1080 pixels vertically).

How Resolution Affects the Display?

- ✓ Higher Resolution: More pixels → Sharper and more detailed images.
 - ✓ Example: A 4K monitor (3840 x 2160) has significantly better detail than a Full HD monitor (1920 x 1080).
- ✓ Lower Resolution: Fewer pixels → Less sharp and more pixelated images.
 - ✓ Example: A 1024 x 768 resolution will appear grainy on a large screen.

Common Monitor Resolutions

Resolution Name	Pixel Dimensions	Aspect Ratio	Use Case
HD (720p)	1280 x 720	16:9	Entry-level monitors, older devices
Full HD (1080p)	1920 x 1080	16:9	Standard for general-purpose use
Quad HD (1440p)	2560 x 1440	16:9	Gaming, mid-range monitors
4K Ultra HD (2160p)	3840 x 2160	16:9	High-end monitors, professional work
8K Ultra HD	7680 x 4320	16:9	Advanced displays, large screens

Aspect Ratio

- **Aspect Ratio :**

- The aspect ratio of a display refers to the proportional relationship between its width and its height.
- It is typically expressed as two numbers separated by a colon, such as 16:9 or 4:3.
- The first number represents the width, and the second number represents the height.
- For example, a 16:9 aspect ratio means that for every 16 units of width, there are 9 units of height.

Refresh Rate

- **Refresh Rate :**

- The refresh rate of a display, measured in Hertz (Hz), indicates how many times per second the image on the screen is redrawn or refreshed.
- A higher refresh rate typically results in smoother motion and reduced motion blur, especially in fast-paced content such as video games or action movies.



Thank You

